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C/EBPα bypasses cell cycle-dependency during immune cell transdifferentiation

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Our earlier work has shown that pre-B cells can be converted into macrophage-like cells by overexpression of the transcription factor C/EBPα or C/EBPβ with high efficiency. Using inducible pre-B cell lines, we have now investigated the role of cell division during C/EBP-induced reprogramming. The majority of cells reprogrammed by C/EBPα incorporated BrdU before arresting at G_0, and all C/EBPβ-induced cells incorporated the compound. This contrasts with reports from other systems where transdifferentiating cells essentially do not divide. Although inhibition of DNA synthesis led to an impairment of C/EBPα-induced transdifferentiation, sorted G_0/G_1 and G_2/M fractions showed no significant differences in their reprogramming kinetics. In addition, knocking-down p53 did not accelerate the transdifferentiation frequency, as it has been described for reprogramming of induced pluripotent (iPS) cells. Time-lapse experiments showed that, after C/EBPα induction, approximately 90% of cells divide once or twice, while 8% do not divide at all before acquiring a macrophage phenotype, supporting our BrdU incorporation results. Importantly, the non-dividing cell subset expressed the highest levels of C/EBPα and was the fastest in differentiating, suggesting that high levels of C/EBPα accelerate both the switching process and the cells' growth arrest. Our data show that traversing the cell cycle is not strictly required for pre-B cell to macrophage conversion and provides new evidence for the notion that the mechanisms of transcription factor induced transdifferentiation and iPS cell reprogramming differ.

Introduction

Transcription factor-induced cell reprogramming has revolutionized the stem cell field. There has been an explosion in the last few years of reports describing direct transdifferentiation from one cell type to another. After the finding that MyoD can convert fibroblasts into muscle cells,[1] numerous other incidences of transdifferentiation have been reported, including that of immature and mature B cells into macrophages,[2] committed T cell precursors into macrophages and dendritic cells,[3] adult pancreatic exocrine cells into β-cells[4] and fibroblasts into neurons,[5] cardiomyocytes[6] and hepatocyte-like cells[7] (see also review in ref. 8). In a separate line of work, Takahashi and Yamanaka have shown that it is possible to reprogram somatic cells into "embryonic stem cells" (so-called induced-pluripotent stem cells, iPSCs) by overexpressing a set of four transcription factors.[9]

Mechanisms that link the cell cycle with differentiation or transdifferentiation are poorly understood and remain somewhat controversial. Cell cycle progression and differentiation can be considered as two distinct and mutually exclusive processes during development: cells that are cycling typically do not differentiate, while cells that have terminally differentiated cease to divide. In the 70s and 80s, Holtzer et al. proposed that terminal differentiation requires a critical cell division ("quantal mitosis").[10] Further, it was suggested that, during DNA replication, remodeling of chromatin allows access of regulatory factors to previously inactive regulatory domains.[10,11] However, chicken myoblasts can be induced to differentiate into multinucleated myotubes in the absence of cell proliferation[12] and BrdU incorporation experiments have demonstrated that the majority of cells do not divide during transcription factor-induced transdifferentiation of pancreatic exocrine into endocrine cells and of fibroblasts into neurons.[4,5] On the other hand, increasing the cell division rate, such as by p53 or p21 knockdown, accelerates reprogramming of fibroblasts and of B cells into iPS cells.[13-15]

Our earlier work has shown that overexpression of the transcription factors C/EBPα and C/EBPβ can induce the conversion of pre-B cells into macrophages.[2] The transdifferentiation induced by C/EBPα represents a direct transition from one cell type to another.[16] C/EBPα plays a dual role in myeloid differentiation and in cell cycle control. On the one hand, ablation experiments in mice demonstrated that it is required for the formation and committment of myelomonocytic cells.[17] On the other hand, C/EBPα inhibits cell growth through a variety of mechanisms: it inhibits the interaction of the two cyclin-dependent kinases, Cdk2 and Cdk4, with cyclins,[18] induces the expression of the Cdk inhibitor p21[19] and represses the activity of the E2F transcription factor.[20]

C/EBPα-induced transdifferentiation of pre-B cells into macrophages constitutes an ideal system to investigate the question

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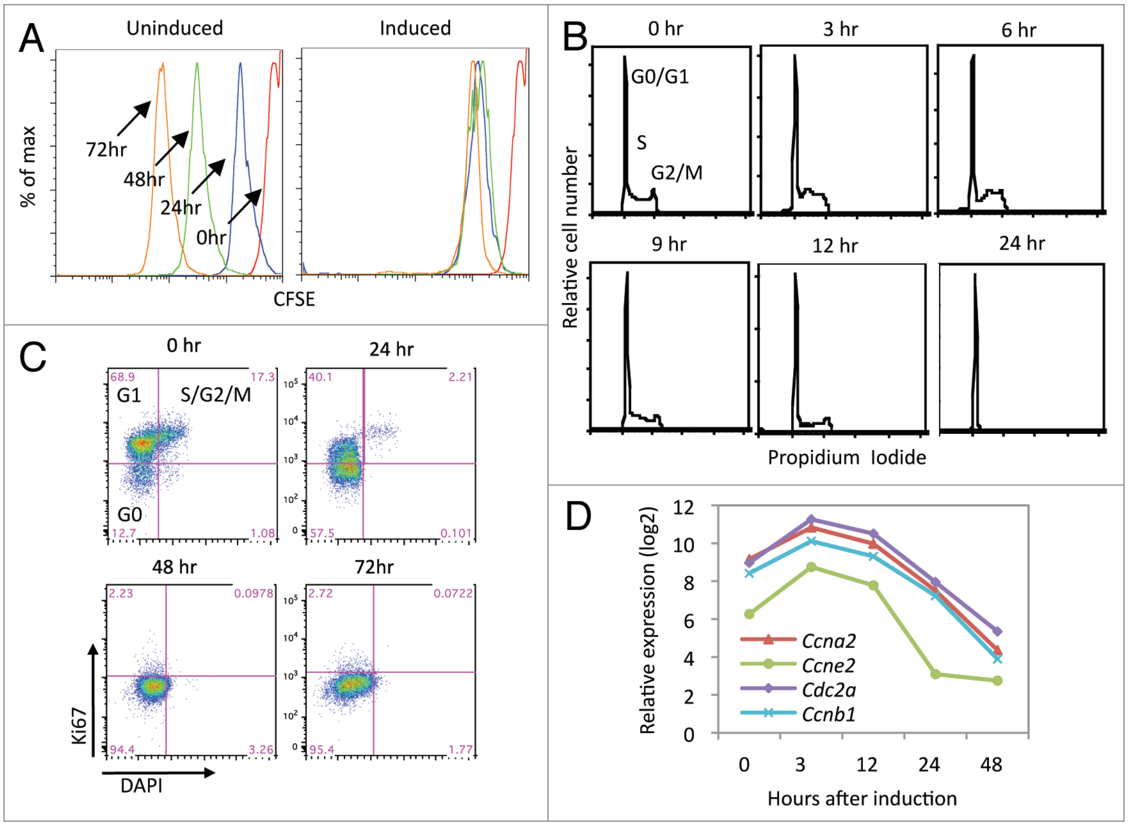

Figure 1. Cell cycle analysis of pre-B cells during C/EBPα-induced transdifferentiation. (A) Mean fluorescence intensity of CFSE of uninduced and C/EBPα-induced cells. The colors of the profiles on the right correspond to those indicated on the left. (B) Propidium iodide staining analysis of pre-B cells induced for different times. (C) FACS profiles of pre-B cells at different times of induction, monitoring Ki67 and DAPI expression. The fraction of cells corresponding to the different cell cycle stages are indicated. (D) Affymetrix array expression profiles of four cell cycle genes that were downregulated during reprogramming.

of whether cell division is necessary for the transdifferentiation process.[21] Using this system, we found that the majority of the cells divide once or twice during reprogramming, but that a small yet significant proportion does not divide. Those cells were found to be the first ones to transdifferentiate and to contain high levels of exogenous C/EBPα. We conclude that cell division is not required for the reprogramming of pre-B cells into macrophages and suggest that the cells that do divide represent a carry-over effect from the rapidly cycling pre-B cells.

Results

C/EBPα induced pre-B cells undergo one cell division in average before arresting in G_0 during transdifferentiation. To investigate the role of cell division during C/EBPα-induced reprogramming of pre-B cells into macrophages, we analyzed a pre-B cell line (C11) expressing C/EBPα-ER. This line can be converted at 100% efficiency into functional macrophage-like cells following treatment with β-estradiol (β-Est).[21] We used carboxyfluorescein diacetate succinimidyl ester (CFSE) staining to monitor cell divisions. When CFSE diffuses into cells, its acetate groups are cleaved by esterases, and the compound becomes fluorescent and impermeable. Since CSFE is very stable, its fluorescence intensity decreases by approximately half with every cell division. As expected, C11 cells stained with CFSE proportionally decreased their fluorescence intensity when maintained without induction (**Fig. 1A**, left), suggesting an approximate doubling time of 11 h. In contrast, when C11 cells were induced with β-Est, the CFSE fluorescence intensity decreased by half within the first 24 h and remained stable thereafter (**Fig. 1A**, right). Therefore, cells divided on average once before they arrested their proliferation after one day, confirming results obtained with another inducible cell line.[21] We next analyzed the cell cycle at different time points during transdifferentiation with propidium iodide (PI) staining. As shown in **Figure 1B**, the cells entered into the G_0/G_1 phase 24 h after induction, consistent with the known inhibitory effect of C/EBPα on cell proliferation. We further determined at which point C/EBPα blocks the cell cycle during reprogramming by using Ki67 staining. Our results show that at 24 h after induction, 40% of the cells were blocked in G_0 and 57% in G_1, but that at 48 h, all cells were in G_0 (**Fig. 1C**). Finally, we analyzed gene expression arrays of transdifferentiating cells obtained in our previous work (GEO number GSE17316).[21] We found that many genes required for cell cycle progression, including *cyclinA* (*Ccna2*), *cyclinE* (*Ccne2*), *CDC2* (*Cdc2a*) and *cyclinB1* (*Ccnb1*), were dramatically downregulated during reprogramming (**Fig. 1D**). The significance of the observed transient upregulation of these markers is unclear. In conclusion, during C/EBPa induced transdifferentiation, pre-B cells divide on average once before arresting in the G_0 stage.

A subset of pre-B cells induced by C/EBPα, but not C/EBPβ, transdifferentiated without exhibiting DNA synthesis. To determine whether all cells induced by C/EBPα pass through the S phase before converting into macrophages, we studied 5-bromodeoxyuridine (BrdU) incorporation. When C11 cells were incubated with both β-Est and BrdU for 24 h and then analyzed by FACS, 90.5% of the cells were found to have incorporated the nucleoside analog, while the rest were negative (**Fig. 2A**). In contrast, and as expected, uninduced C11 cells scored 100% positive for BrdU incorporation (**Fig. 2A**). Our earlier work showed that C/EBPβ, a transcription factor closely related to C/EBPα, is also capable of inducing pre-B cell transdifferentiation, but without inducing withdrawal from the cell cycle.[2] To test the relationship between C/EBPβ-induced transdifferentiation and the cell cycle, we used a C11 cell line derivative (C11α/β) that contains, in addition to C/EBPαER, a doxycycline-inducible form of C/EBPβ. CSFE-labeling experiments showed that the growth of C11α/β cells was arrested 2 d after treatment with 1 μg/μl doxycycline, at which time they partially upregulated Mac-1 and downregulated CD19 (**Fig. S1**). BrdU incorporation showed that 100% of the C/EBPβ cells went through S phase (**Fig. 2B**). Together, these results show that the vast majority of pre-B cells induced to differentiate by C/EBPα and C/EBPβ go through S phase, raising the possibility that transdifferentiation requires cell division.

Inhibition of cell division blocks establishment of the myeloid program. Next, we tested the effect of cell cycle inhibitors on transdifferentiation. We used aphidicolin as a highly specific DNA synthesis inhibitor that blocks DNA polymerase α. Since we found that effective concentrations induced cell death, we decided to create a C11 derivative that overexpresses the anti-apoptotic gene *Bcl2* (C11-Bcl2). Treating these cells with 20μM aphidicolin for 48 h caused an almost complete block of cell proliferation, as judged by CFSE staining (**Fig. 3A**). We therefore pre-incubated the cells for 48 h with aphidicolin and then induced them with β-Est or kept them uninduced while maintaining the treatment with the DNA synthesis inhibitor. Subsequent FACS analyses showed that Mac-1 upregulation was inhibited in the presence of aphidicolin, but that CD19 downregulation was not affected or even slightly accelerated after induction (**Fig. 3B**; **Fig. S3C**). The inhibition of the macrophage program could also be detected by a lack of morphological changes, an inhibition of cell size and of granularity characteristic of macrophages. Furthermore, in the presence of aphidicolin, the phagocytic capacity was reduced from > 90% in the control Mac1+/CD19- cells to about 30% (**Fig. 3C**). Together, these results suggest that the bulk of pre-B cells induced by C/EBPα need to traverse the cell cycle to effectively activate the macrophage program, whereas extinction of the B cell program, as determined by CD19 expression, is not diminished in the absence of DNA synthesis.

The cell cycle stage of pre-B cells does not influence the reprogramming kinetics. The above findings suggested that DNA synthesis or another cell cycle checkpoint is required for

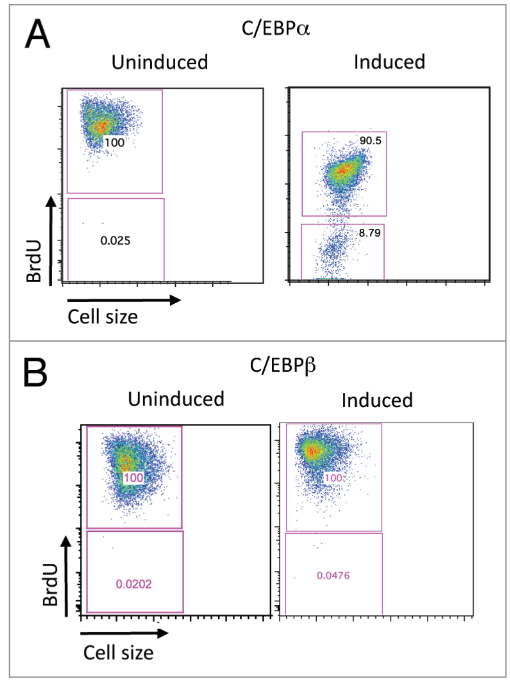

Figure 2. DNA synthesis during C/EBPα- and C/EBPβ-induced pre-B cell to macrophage conversion. (A) FACS profiles of BrdU incorporation in uninduced C11 cells and cells 24 h after induction with C/EBPα. (B) FACS profiles of BrdU incorporation in uninduced C11 cells and cells 24 h after induction with C/EBPβ.

transdifferentiation. To test this possibility further, we first asked whether transdifferentiation kinetics is influenced by the cell cycle stage of the starting population. The idea was that if transdifferentiation of pre-B cells into macrophages is dependent on cell cycle, then reprogramming kinetics of the two phases should be different, with G_0/G_1 cells reprogramming being substantially faster than G_2/M cells, since these cells first have to go through M and G_1 before reaching S phase (**Fig. 4A**). To sort C11 cells into G_0/G_1 and G_2/M cell subsets, we used staining with Vybrant Ruby (**Fig. 4B**), a non-toxic DNA dye used for cell cycle analyses in living cells. Re-analysis of sorted fractions confirmed that the two populations were well separated (**Fig. 4C**). Cells were then induced with β-Est, in parallel to unsorted control C11 cells. We found that the transdifferentiation kinetics, as determined by downregulation of CD19 and upregulation of Mac1, were indistinguishable between cells in G_0/G_1 phase (**Fig. 4D**, top) and those in G_2/M phase (**Fig. 4D**, bottom).

Our earlier work showed that a pulse induction of 12 h on C11 pre-B cells, followed by washout of the inducer, was sufficient to convert a significant proportion of cells into macrophages.[21] We therefore repeated the above experiments using a 12 h β-Est pulse induction, instead of a continuous induction protocol, on the two sorted cell cycle stages. This assured that most of the induced cells have gone through one cell cycle, and that exogenous C/EBPα did not cause a continuous inductive pressure. However, even under these conditions, the kinetics of Mac1 upregulation were found to be indistinguishable between the G_0/G_1 and the G_2/M fractions (**Fig. S2A**).

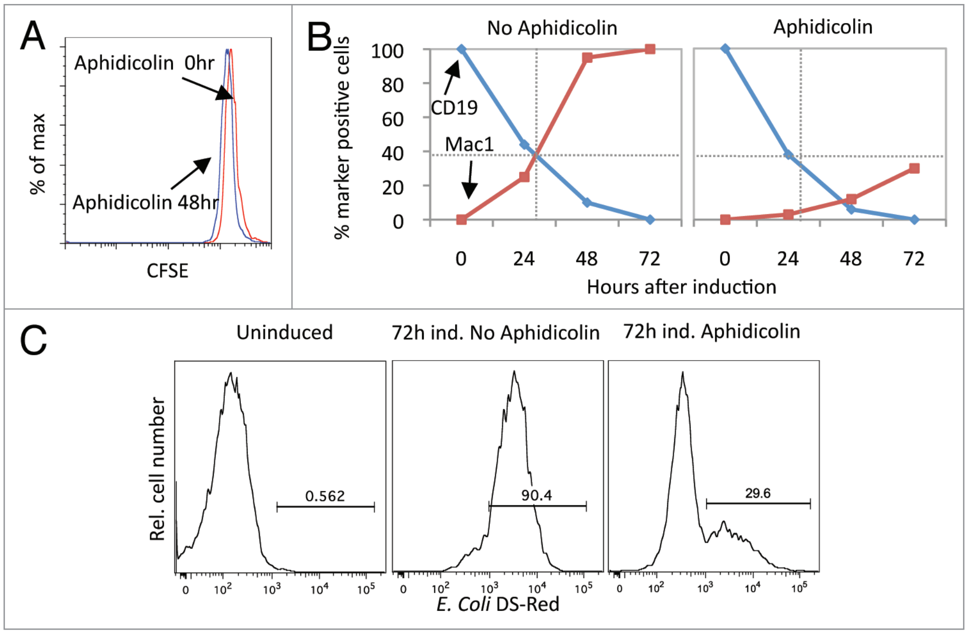

Figure 3. Effect of aphidicolin on the transdifferentiation of pre-B cells. (A) CFSE fluorescence intensity of uninduced C11-Bcl2 cells treated with the DNA synthesis inhibitor aphidicolin showing absence of cell division after 48 h. (B) Kinetics of CD19 downregulation and Mac1 upregulation during reprogramming of pre-B cells in the absence and presence of aphidicolin. (C) Effect of aphidicolin on the phagocytic capacity of induced C11-Bcl2 cells. Uninduced cells and cells induced for 72 h were grown in the presence or absence of aphidicolin and incubated with *E. coli* expressing dsRed prior to FACS analysis.

showed that when pre-B cells transdifferentiate into macrophages, they become irregular in shape and are highly motile as early as 15 h after induction.[21] To address the question of whether all cells divide during C/EBPα-induced reprogramming, we therefore performed time-lapse experiments of C11 cells after induction with β-Est. We found that 5.8% of the induced pre-B cells divided twice, 86.4% divided once, and 7.8% of did not divide at all before becoming motile and changing their morphology (**Fig. 5A**; **Fig. S3A**). Importantly, there were significant differences in the timing of transdifferentiation between these three groups of cells: cells that did not divide transdifferentiated into macrophage-like cells as early as after 8 h, while cells that divided once transdifferentiated first after 17 h, and cells that divided twice, after 23 h (**Fig. 5B**). It also became apparent that in most cases, the two daughter cells transdifferentiated at about the same time. These results confirm the finding that a small subset of induced C11 cells do not incorporate BrdU, supporting the notion that cell divisions are not strictly required for reprogramming.

Several groups have shown that p53 ablation accelerates the kinetics of induced pluripotent stem (iPS) cell formation, and that this acceleration is directly proportional to the increase in cell proliferation caused by inhibition of the tumor suppressor gene.[13-15,22] We therefore tested whether a p53 short hairpin RNA (shp53) likewise accelerates the kinetics of B cell to macrophage transdifferentiation in our system. To this end, we infected C11 cells with a lentivirus containing shp53 and created a stable cell line (C11-shp53) in which the levels of p53 mRNA were decreased by 60% (**Fig. S2B**). C11-shp53 cells treated for 2 h with BrdU showed almost a 3-fold increase in the incorporation of the nucleoside analog compared with C11 cells (**Fig. S2C**). Nevertheless, after inducing transdifferentiation, we found no acceleration of Mac1 upregulation or CD19 downregulation (**Fig. 4E**). We also tested whether cell status changed during reprogramming of C11-shp53 cells using Ki67 antibody staining. Two days after induction, all the cells were in G_0 (**Fig. S2D**), exactly like control cells without shp53. We therefore conclude that, unlike reprogramming of somatic cells into iPS cells, ablation of p53 does not accelerate the conversion of pre-B cells into macrophages, nor does it prevent cell cycle arrest induced by C/EBPα. Taken together, these results indicate that neither G_1/S nor G_2/M transitions represent essential checkpoints for reprogramming.

Time-lapse experiments reveal a subset of rapidly transdifferentiating cells that do not require cell division. To investigate the apparent discrepancy that DNA synthesis inhibitors can impair transdifferentiation, yet traversing the S phase is not required for induced transdifferentiation, we first performed time-lapse experiments. Our earlier work with this technique

Pre-B cells that transdifferentiate without cell division express the highest C/EBPα levels. Previous work has shown that high levels of C/EBPα induce the transdifferentiation of a higher percentage of primary pre-B cells than lower concentrations of the factor.[2] Therefore, the observed positive correlation between the speed of transdifferentiation and of the proportion of cells that did not divide raised the possibility that the most rapidly switching cells contained higher levels of C/EBPα. To test this, we performed a BrdU incorporation experiment with C11 cells induced with β-Est and determined the relative C/EBPα levels, as indicated by the fluorescence intensity of GFP, by gating BrdU-negative (R1) and -positive cells (R2) (**Fig. 5C**, left). BrdU-negative cells were significantly enriched in the GFP-high fraction (**Fig. 5C**, right). In addition, the fraction of cells that did not incorporate BrdU increased from 3% to 28% in C/EBPα-low and -high cells, respectively (**Fig. S3C**). These results indicate that high levels of C/EBPα inhibit cell division more effectively than lower levels. Next, we tested whether the rate of Mac-1 upregulation was affected by C/EBPα levels. Indeed, as shown in **Figure 5D** (left), cells with high amounts of C/EBPα upregulated Mac-1 expression much more rapidly than C/EBPα-medium or -low cells. In contrast, the rate of CD19 downregulation was essentially not affected (**Fig. 5D**, right). Similar findings were also made with C/EBPβ-induced cells (**Fig. S4**). These experiments therefore reveal that the transdifferention of pre-B

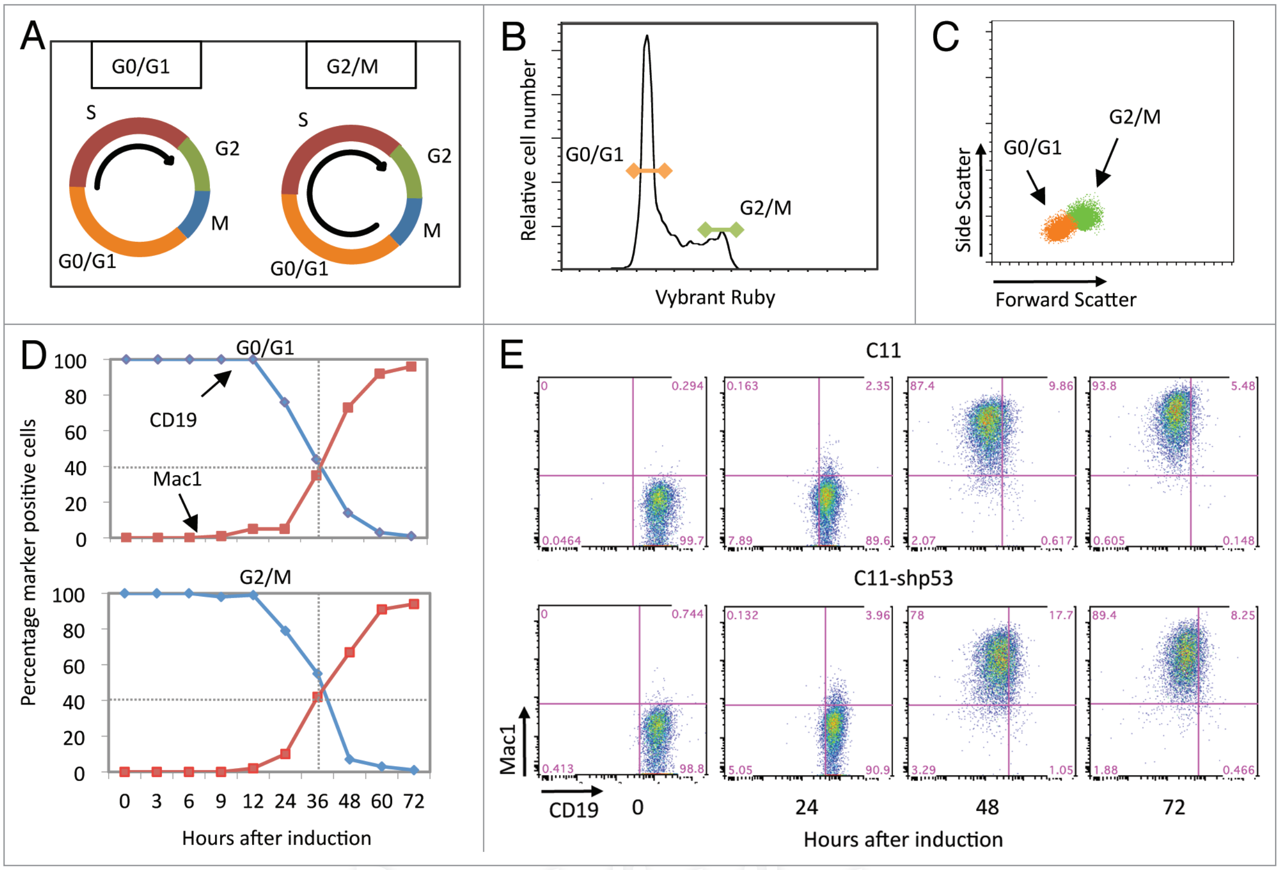

Figure 4. Transdifferentiation kinetics of G_0/G_1 and G_2/M fractions from inducible pre-B cells. (A) Diagram of cell cycle position after sorting of G_0/G_1 (left) and G_2/M (right) fractions, with curved arrows indicating trajectories necessary to traverse S phase. (B) Cell cycle profile of pre-B cells using Vybrant Ruby staining. The colored lines indicate the fractions selected for cell sorting. (C) Re-analysis of the sorted fractions, with the obtained FACS plots overlaid. (D) Kinetics of reprogramming of G_0/G_1 phase and G_2/M phase fractions, monitoring CD19 and Mac1 expression. The dotted lines represent the crossing points of the CD19/Mac1 kinetics of the G0/G1 fraction. (E) FACS profiles of Mac1 and CD19 expression during reprogramming of pre-B cells in absence (top) or presence (bottom) of a short hairpin RNA against the p53 gene.

cells to macrophages is accelerated by high levels of C/EBPα/β, and that in the case of C/EBPα, this leads to a decrease in the proportion of the cells that traverse the cycle.

Discussion

Our results have shown that during C/EBPα-induced transdifferentiation of pre-B cells into macrophages the majority of cells divide at least once, and that blocking DNA synthesis dramatically impairs transdifferentiation. However, the rate of transdifferentiation was independent of whether the starting cells were in G_0/G_1 or G_2/M and a knockdown of p53, a treatment that accelerated the cells' growth rate, had no detectable effect on the transdifferentiation kinetics. In addition, approximately 10% of cells did not divide, as shown by BrdU incorporation and time-lapse experiments, indicating that cell division is not strictly required for reprogramming in this system. The finding that the non-dividing cells correspond to the most rapidly transdifferentiating cell subset, and that this subset expresses the highest levels of C/EBPα, suggests that they were forced to withdraw from the cell cycle by C/EBPα. These conclusions are indirectly supported by the findings with the transcription factor C/EBPβ: here, pre-B cells converted into macrophages show a delay in cell cycle withdrawal compared with C/EBPα. Therefore, as expected, all transdifferentiating cells incorporate BrdU.

Our observation that immune cell transdifferentiation requires no cell division is in agreement with the finding that there are no significant promoter DNA methylation changes during transdifferentiation of pre-B cells into macrophages, with macrophages essentially maintaining the overall methylation pattern seen in pre-B cells.[23] This suggests that no major changes in DNA methylation are required for cell switching, although it would be interesting to know whether inhibition of DNA methylation modulates the switching rate. In apparent contrast to transdifferentiation, DNA demethylation appears to play a critical role during iPS cell reprogramming,[24-26] although it is controversial whether this requires DNA synthesis. Our findings also broadly confirm and extend conclusions reached in studies where fibroblasts were converted into neurons and pancreatic exocrine into endocrine cells.[4,5] The major difference to our results is that, in these systems, the vast majority of cells do not incorporate BrdU during transdifferentiation. A plausible interpretation is that only a low proportion of primary fibroblasts and endocrine cells incorporate the compound, and so the induced conversions represent

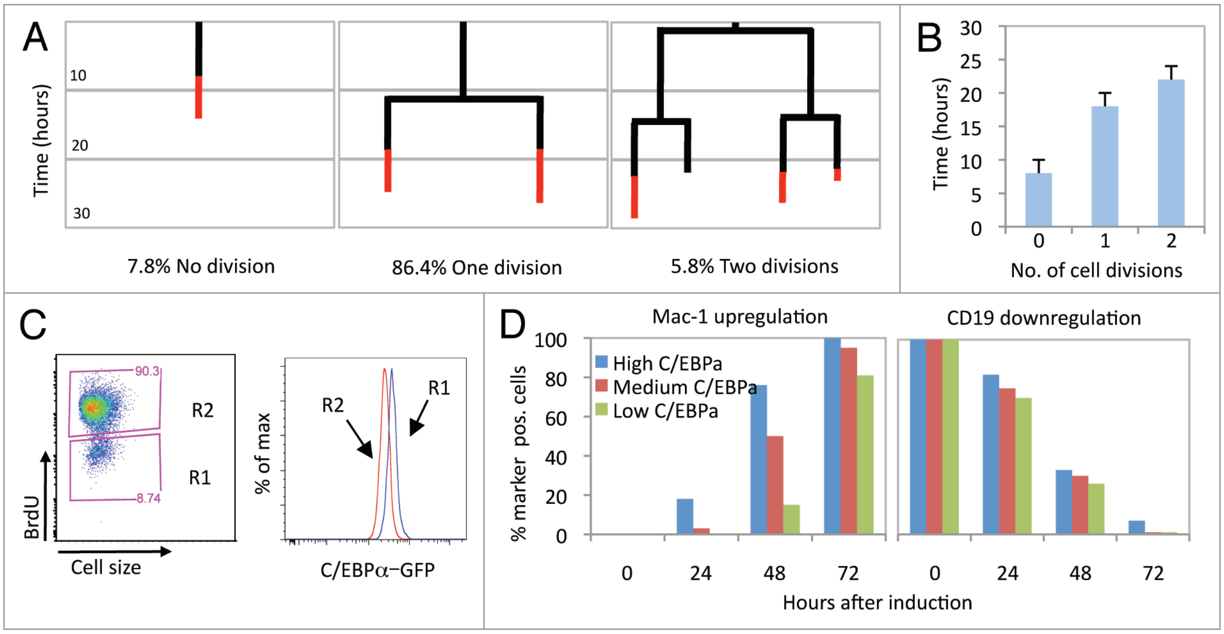

Figure 5. Time-lapse experiments and effects of C/EBPα dosage on transdifferentiation of pre-B cells. (A) Time-lapse microscopy analysis of transdifferentiating cells on a time scale from top to bottom. Results are shown from individual cells that did not divide (left), divided once (middle) or divided twice (right). The black bars indicate no differentiation, and the red bars differentiation as determined by increased motility and acquisition of an irregular cell shape. The percentage of cells in each category is indicated. (B) Correlation of the time required for differentiating with the number of cell divisions. (C) FACS profiles monitoring BrdU incorporation in C11 cells 24 h after induction (R1 and R2, BrdU negative and positive cells, respectively, on the left), with C/EBPα-GFP expression in R1 vs. R2. (D) Kinetics of reprogramming at different times of induction. Gated cells with high, medium and low levels of C/EBPα, as monitored by Mac-1 upregulation (left) and CD19 downregulation, are shown (right).

transitions between two essentially quiescent cell populations. In contrast, the switching system studied here represents a transition between rapidly proliferating cells (C11 pre-B cells divide approximately every 11 h) and macrophages arrested in G_0. It therefore appears that during C/EBPα-induced transdifferentiation, the lymphoid cells exhibit a "carry over" effect, consisting of one or two rounds of cell division before turning into quiescent macrophages.

The observation that the DNA synthesis inhibitor aphidicolin impaired C/EBPα-induced transdifferentiation was unexpected and is in apparent conflict with our finding that the S phase itself does not appear to represent a checkpoint required for myeloid gene activation. A possible explanation is that C/EBPα-induced cells prevented from cycling accumulate an unknown inhibitor of a myeloid regulator or fail to accumulate sufficient levels of a hypothetical co-factor.

The fact that the C/EBPα-induced cells were not prevented by shp53 from arresting in GO suggests that the cell cycle inhibitory effect of C/EBPα is dominant over the growth accelerating effect of shp53. Whether this plays out at the level of transcriptional regulation of the cell cycle inhibitor p21, a direct target of both C/EBPα and p53,[19,27] remains to be determined. Our observation that a p53 knockdown does not alter the transdifferentiation kinetics of pre-B cells into macrophages contrasts with its accelerating effects on iPS cell reprogramming (14–16). This difference lends further support to the postulate[28] that the mechanisms of transcription factor-induced transdifferentiation and iPS cell reprogramming are fundamentally different.

Methods

Cells and viral constructs. The construction of MSCV C/EBPαER IRES hCD4 virus and the creation of the C11 cell line was as described.[21] The Bcl2 cDNA was taken from pSFFV-Bcl2 (Addgene) and cloned into the pMSCV-puro with EcoRI digestion. The cell line C11-Bcl2 was generated by infecting C11 cells with the pMSCV-Bcl2-Puro retrovirus, selecting with puromycin and sorting single cells. The shp53 lentivector was a kind gift of Dr Bill Keyes. The cell line C11-shp53 was generated by infecting C11 cells with the pMSCV-shp53-Puro-GFP lentivirus, selecting with puromycin and sorting single cells. The pHAGE-tetO-C/EBPβ-IRES-tdTomato virus was constructed by inserting C/EBPβ cDNA in the pHAGE vector, kindly provided by Dr. Gustavo Mostoslavsky,[29] using NotI and BamHI. To generate the C11α/β line, C11 cells containing rtTA were infected with tetO-C/EBPβ tdTomato virus, and a single cell-derived clone selected.

Cell reprogramming, cell cycle inhibition and FACS analyses. C11 cell lines were induced with 100 nM β-Est (Calbiochem) and grown in special induction medium containing IL-3 and mCSF-1 (10 ng/mL) (Peprotech). The C11α/β cell line was induced with 1 μg/μL doxycyclin (Sigma). For the pulse induction experiment, cells were thoroughly washed and then incubated with 10 μM of the β-Est antagonist ICI (Tocris Bioscience). For the cell cycle inhibition experiments, cells were treated for 2 d with 20 μM aphidicolin (Calbiochem). Antibodies to cell surface antigens were purchased (BD PharMingen). Cells were analyzed with a FACS LSRII flow cytometer (BD Biosciences) using FlowJo software (Tree Star) and sorted with a FACS ARIA flow cytometer (BD Biosciences).

Real-time PCR. qRT-PCR reactions were performed in triplicate as described.[21] Ct values were normalized to glucuronidase β (GusB), and the relative expression was calculated by the Pfaffl method.[30]

Cell cycle analysis. Cell cycle analyses were performed with the following different techniques. For CFSE staining, 3 x 10^5 cells were resuspended in 1 mL PBS with 0.2 μM CFSE for 5 min (Invitrogen) 4 h before the induction and then rinsed twice with 4 volumes of PBS. For propidium iodide staining, 2 x 10^5 cells were resuspended in 1 mL of cold 70% ethanol, stored overnight at 4°C, and then rinsed with PBS and stained for 15 min on ice with 300 μL cell cycle solution of 50 μg/mL PI (Sigma), 0.1 mg/mL RNaseA (Sigma) in PBS. For Ki67 staining, 2 x 10^5 cells were resuspended in 100 μl of Fix and Perm Medium A (Invitrogen), incubated for 15 min at room temperature and rinsed with PBS. This was then resuspended in 100 μL of Fix and Perm Medium B (Invitrogen) with 5 μL of anti-Ki67-PE (BD Biosciences), incubated for 20 min at room temperature, rinsed with PBS, resuspended in 500 μL of PBS containing 5 μg/mL RNaseA and 2 μg/mL DAPI, incubated for 15 min at room temperature and FACS analyzed. For BrdU staining, cells were treated for 24 h with 50 μM BrdU (2 x 10^5 cells in 1 mL of growth medium), resuspended in 100 μL of Fix and Perm Medium A (Invitrogen), incubated for 15 min at room temperature, rinsed with PBS and resuspended in 100 μL of Fix and Perm Medium B (Invitrogen). Following another 20 min incubation at room temperature, cells were rinsed with PBS, resuspended in 100 μL of PBS with 10 μL of 10 DNaseI buffer (100 mM Tris-HCL, 25 mM $MgCl_2$, 5 mM $CaCl_2$) and 2 μL of DNase I solution (Promega 22U/μL) and incubated 1 h at 37°C in the dark. They were then resuspended in 100 μL PBS with 5 μL anti-BrdU-PerCP-Cy5.5 (BD Biosciences), rinsed and FACS analyzed. For Vybrant Ruby staining, 2 10^6 cells were resuspended in 2 mL of growth medium with 4 mL of Vybrant Ruby (Invitrogen), incubated 1 h at 37°C in the dark and centrifuged; cells were kept in 1 mL of the same solution in which they were stained.

Time-lapse. Microscopy chamber wells (μ-Slide 8 well coated poly-l-lysine from IBIDI) were pre-treated with 200 μL retronectin (Takara) (48 μg/mL) for 2 h at room temperature. Retronectin was removed and 200 μL PBS, 2% BSA was added for 30 min. Wells were then rinsed with 300 μL PBS and left overnight in the incubator. 6 x 10^3 cells were resuspended in 200 μL of growth medium (with or without β-Est, IL3 and mCSF1) and added to the wells. Pictures were taken every 5 min (using four positions per well).

Phagocytosis assay. C11-Bcl2 cells were seeded into 6-well plates, treated with 20 μM aphidicolin for 2 d, and then induced with β-Est for 3 d. 100 dsRed *E. coli* per cell were added, and plates were centrifuged at 800 g for 15 min. Thereafter, cultures were incubated with 400 μg/mL gentamycin for 3 h at 37°C to eliminate extracellular bacteria. To remove excess bacteria from cells, a 2-mL suspension was underlayered with 5 mL of fetal calf serum and kept at room temperature for 2 h. Cells were collected from the serum phase, centrifuged and analyzed by FACS.

Disclosure of Potential Conflicts of Interest

No potential conflicts of interest were disclosed.

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Supplemental Material

Supplemental materials may be found here:
www.landesbioscience.com/admin/article/21119

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